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Water Stress in India

Background

India's population exceeds over 1.4 billion, making it the most populous country in the world. With its vast geography and climate, it is rich in culture and diversity. Although India continues to grow its economy and population, it has also grown in its water scarcity and poor water sanitation. According to Water.org , about 342 million of India's population lack access to clean and safe water. The lack of access to safe and clean water is directly correlated to the depleting groundwater, climate change, and rising sea levels.

The purpose of this project is to explore a data set highlighting India's climate and geography, and see if it could provide enough insight to someone trying to predict the water stress index of a region, and what the best predictors would be.

Data

The data used for this project comes from Kaggle.com called "India's Agricultural & Environmental Indicators." It consists of 36 states or unions in India, each with a diversely unique climate and geographical factors. Along with that, are measurements of their precipitation, winter and summer temperatures, irrigation coverage, groundwater availability, cropping intensity (the ratio of cropped area to net sown area), and level of water stress. It also provides the approximate longitude and latitude of each state/union.

For predicting water stress index, the only variables that are considered are latitude, longitude, precipitation, and the average temperatures, as these are the best parameters that consider both the geographical and atmospheric effects of the particular state or region.

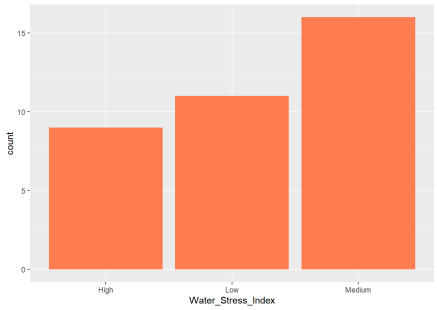
Variable	Range or types
Soil type	Red and black, mountain, alluvial, red and laterite, laterite, black and alluvial, black, red and yellow, laterite and red, arid/desert, red, laterite and alluvial, lateritic, cold desert, coral and sandy, coastal alluvial

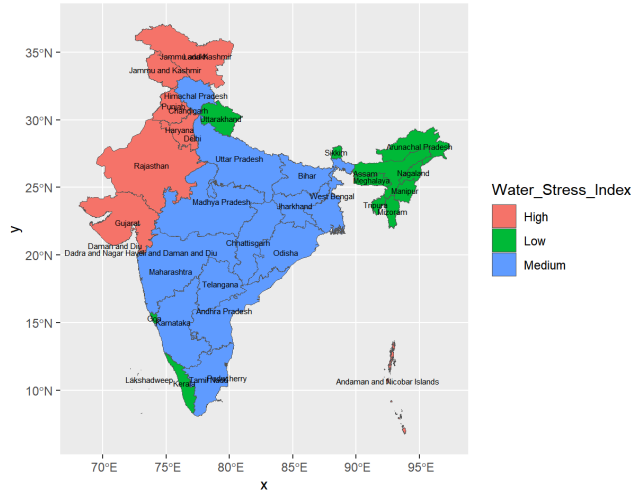
Geographical region	Eastern ghats/coastal plains, eastern himalayas, brahmaputra valley, brahmaputra valley, indo-gangetic plains, central india, + 23 more
Climate zone	Tropical wet, tropical wet-dry, humid subtropical, subtropical, tropical monsoon, arid/tropical wet_dry, humid subtropical/alipine, subtropical semi-arid, temperate/alipine, cold desert/alpine, tropical maritime, arid/desert
Latitude	[10.5667, 341526]
Longitude	[71.1924, 94.7278]
Precipitation (mm)	[100, 3500]
Average summer temp (C)	[22, 38]
Average winter temp (C)	[5, 20]
Irrigation coverage	[20,80]
Groundwater availability	High, medium, low
Cropping intensity	[110,170]
Water stress index	High, medium, low

Exploratory Data Analysis

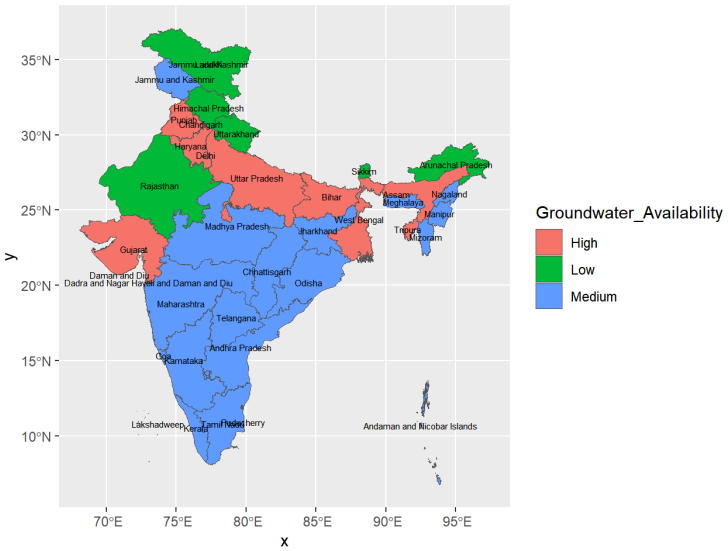
Before making any prediction models, it's best to see the spread and frequency of that data through bar graphs, histograms, or scatterplots.

In an ideal world, water stress is at a low. First looking at the frequency of the water stress index, it is noticeable that most of India has a medium stress index. The bar graph gives false hope since it shows that the amount of states/regions that have a high stress index is the smallest. But a further look into the map of the water stress will show that the total area of India with a small water stress index is smaller than the area covered by high stress index. Those who have a very high index are mostly located in the north-western region of India, which borders Pakistan.

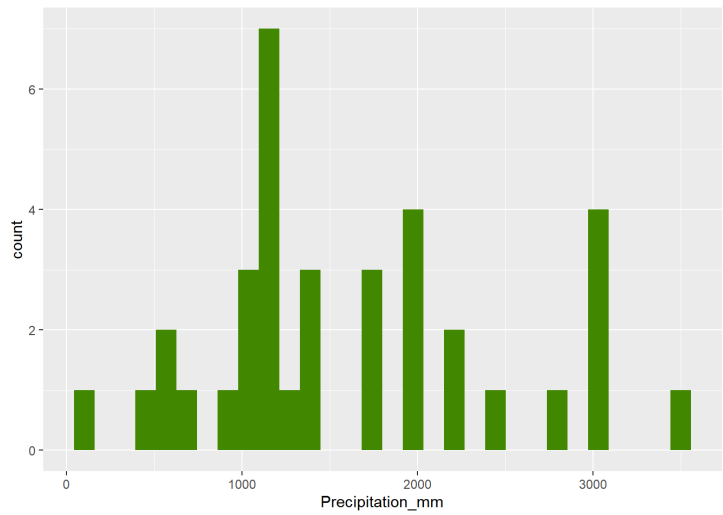




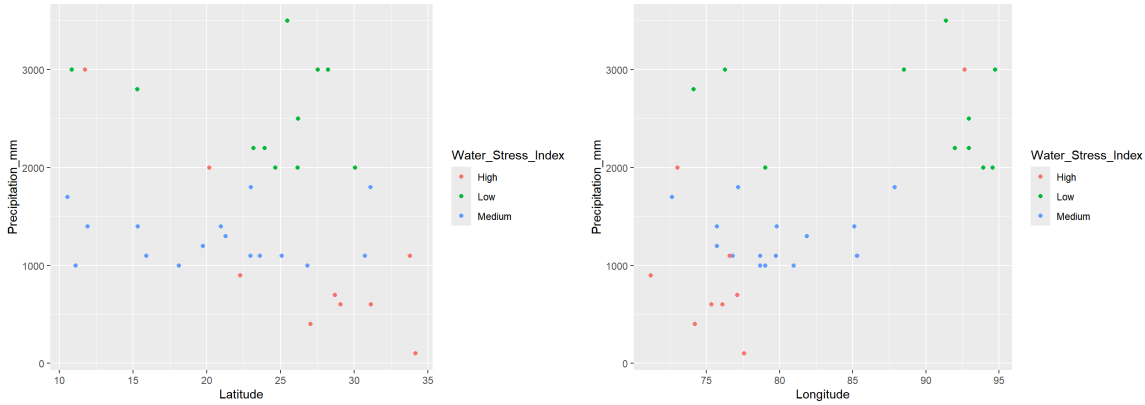
It is difficult to see the correlation of groundwater availability and water stress since they both go by 3 levels (high, medium, low). By looking at the map of groundwater availability, there is some overlap between places with low groundwater and high water stress. Also, a major part of the medium water stress index is shared with the medium groundwater availability. This hints that the availability of groundwater directly affects water scarcity.



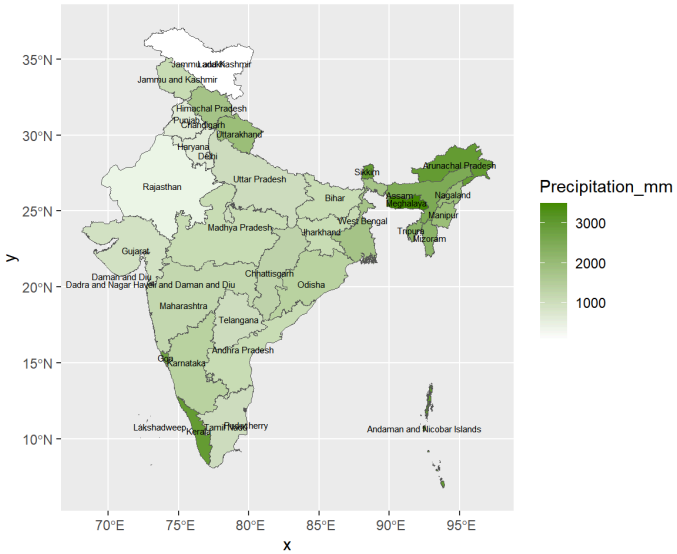
The next step is to compare the amount of precipitation to the water stress index. The distribution of the amount of precipitation is right tail distributed with its peak at 1,200mm.



Looking at a scatterplot between precipitation and latitude, there doesn't seem to be much of a correlation, but there seems to be a positive correlation between precipitation and longitude. Also, it can be seen that the higher the precipitation, the lower the stress water index is.



Lastly, comparing the map of precipitation to the map of water stress, the areas of higher precipitation match the areas of low stress, while the areas with low precipitation align with the high stress regions.



Predicting

With the relationships of the variables in mind, I'd like to see which combination of variables (longitude, latitude, precipitation, or temperature) will give the best predictions of the water stress index. First, the data was split in half into two random samples, one for training and the other for testing. The training data will be fit to a multinomially logistic regression model and then used to predict the level of stress. Those values will then be compared to the true stress index, which can then be used to find the accuracy of the model.

Fitting the model to precipitation, average summer temperature, average winter temperatures, longitude, and latitude gives an accuracy of 0.722 as five out of the eighteen test indexes were misassigned.

```
p        High Low Medium
  High      2   0      1
  Low       0   5      0
  Medium    3   1      6
[1] 0.7222222
```

Fitting with all of the previous except longitude, there is only one index misassigned, giving an accuracy of 0.944.

```
p        High Low Medium
  High      4   0      0
  Low       1   6      0
  Medium    0   0      7
[1] 0.9444444
```

Using all the predictors but latitude yields the same result as the previous model.

```
p        High Low Medium
  High      4   0      0
  Low       1   6      0
  Medium    0   0      7
[1] 0.9444444
```

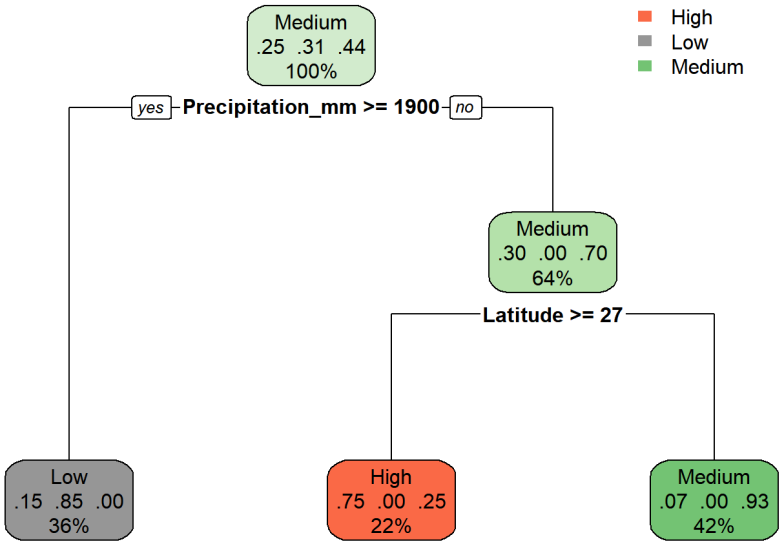
It seems the model runs best when either longitude or latitude are excluded. Next, the model was tested with both excluded. This led to 2 misclassifications and an accuracy of 0.889.

```
p        High Low Medium
  High      4   0      0
  Low       1   5      0
  Medium    0   1      7
[1] 0.8888889
```

Removing the latitude, longitude, and precipitation from the model yields a 0.556 accuracy with 8 misclassifications. It is starting to look like precipitation is a big predictor in the water level in India.

```
p        High Low Medium
  High      2   0      0
  Low       1   1      0
  Medium    2   5      7
[1] 0.5555556
```

Lastly, a decision tree was made to see if water stress index could be mapped into a flow chart and used for future predicting. Unfortunately, I was unable to use the tree with the training data, so it could not be used for predicting. But it still captured interesting insight about how one might predict water stress index. It indicated that only two predictors, precipitation and latitude, are needed to decide whether the stress index is high, medium, or low. If precipitation is over or equal to 1,900mm, the index is low. If it is over 1,900mm then one must consider the latitude. If latitude is greater than or equal to 27, then the stress index would be high. And if the latitude is lower than 27, the stress index is medium.



Conclusion

This particular data set provides a lot of insight about the factors behind India's water stress index and was great for mapping and modeling. The 'mapindia' package in R also provided a great map to implement the data from "India's Agricultural & Environmental Indicators" and made it much easier to visualize where the water scarcity or climate was impacting.

In conclusion, the highest accuracy was achieved when precipitation, average summer temperature, and average winter temperature were the predictor variables paired with either latitude or longitude. And the maps showed that groundwater availability and precipitation levels also directly correlate to the water stress index.

Sources

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